

Balancing Weights

CML Week 3

2026-09-15

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. **Balancing weights**
4. Double robustness, Double ML and TMLE
5. Heterogeneous Treatment Effects
6. HTE II: Neural Networks
7. Off-policy evaluation and optimization
8. Experimental design
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference

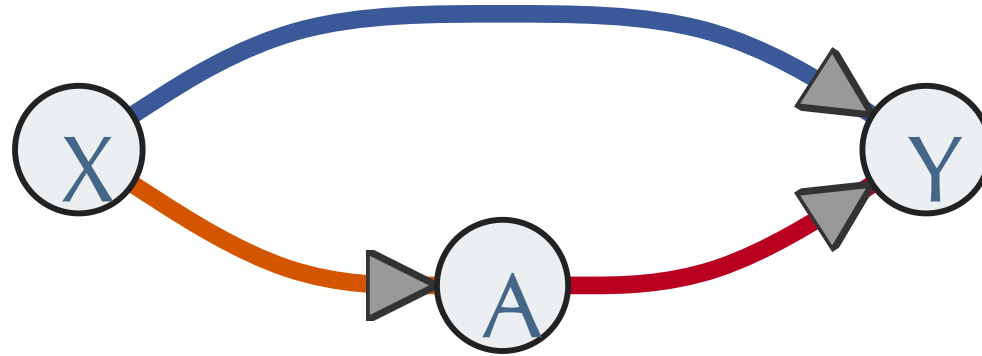
Zooming out

- › Why are we reading methods papers? They're very hard!
- › Look over the syllabus, thinking about the publication dates.
- › The field has advanced a lot in five years.
- › Keep sharp by being able to read into new, difficult methods and topics.

Thus far

- › Metaphysics of Causal Inference
 - » How this is translated into how we think about uncertainty.
 - » Our variance is typically *overestimated* for the simplest target, the SATE.
- › Covariate adjustment in RCTs
 - » It's very useful to reduce variance!
 - » But if we aren't careful we can shoot ourselves in the foot.

This week



- › The edge from $A \rightarrow Y$ is the causal effect we want to measure
- › The edge from $X \rightarrow Y$ is what we're correcting for when we do covariate adjustment
- › The edge from $X \rightarrow A$ is what we're correcting for when we do balancing
- › So what is balancing?

(Rosenbaum and Rubin 1983)

The big picture

- › **The Problem:** Units who receive treatment may systematically differ from those who don't.
- › **The Innovation:** The propensity score $e(x) = P(A = 1 \mid X = x)$ captures all the information in X needed for causal identification.
- › **The Result:** If we condition on the propensity score, treatment assignment is “as good as random” within strata.
- › **Why it matters:** We can reduce a high-dimensional covariate adjustment problem to a single-dimensional one.



Proof sketch of the propensity score theorem

Goal: Show that $(Y(0), Y(1)) \perp A \mid e(X)$

1. Start with unconfoundedness assumption: $(Y(0), Y(1)) \perp A \mid X$

2. By definition of propensity score:

$$e(X) = E[A \mid X]$$

3. $A \perp X \mid e(X)$: *Proof:* For any function $h(X)$, we have:

$$\begin{aligned} E[A \cdot h(X) \mid e(X)] &= E[E[A \cdot h(X) \mid X] \mid e(X)] \\ &= E[h(X) \cdot E[A \mid X] \mid e(X)] \\ &= E[h(X) \cdot e(X) \mid e(X)] \\ &= e(X) \cdot E[h(X) \mid e(X)] \end{aligned}$$

This equality for all functions h is precisely the definition of conditional independence.



Proof sketch of the propensity score theorem (cont.)

4. Combine: $(Y(0), Y(1)) \perp A \mid X$ and $A \perp X \mid e(X)$ implies $(Y(0), Y(1)) \perp A \mid e(X)$

- › If treatment assignment only depends on covariates through the propensity score, then controlling for the propensity score is as good as controlling for all covariates.
- › **Intuition:** The propensity score “balances” the distribution of X across A .



Simple example

Setup: Job training study where treatment was randomly assigned within education levels.

Education strata:

- › High school: 60% get training
- › College: 40% get training
- › Graduate: 20% get training

Key fact: Within each education level, treatment assignment is random.

Propensity score: $e(x) = P(\text{training} = 1 \mid \text{education} = x)$

Result: Conditioning on propensity score achieves the same identification as conditioning on education itself. The propensity score “summarizes” all the relevant confounding information in education.

Balance via weighting

The Horvitz-Thompson approach: Instead of stratifying, we can weight observations to achieve balance.

Inverse propensity weighting (IPW):

- › Weight treated units by $\frac{1}{e(x_i)}$
- › Weight control units by $\frac{1}{1-e(x_i)}$

Intuition: Up-weight units that are rare in their treatment group.

- › If high-education units rarely get training, up-weight the few that do
- › If low-education units rarely stay in control, up-weight the few that do

Result: Creates a “pseudo-population” where treatment is independent of covariates.

ATE estimator: $\hat{\tau} = \frac{1}{n} \sum_i \left(\frac{A_i Y_i}{e(x_i)} - \frac{(1-A_i) Y_i}{1-e(x_i)} \right)$



Bonus: ([Aronow et al. 2016](#))

The Critique: Nonparametric identification (like unconfoundedness) is not enough for credible causal inference.

The Problem: Identification tells you *what* to estimate, $E[E[Y | A = 1, X]]$ or $E\left[\frac{AY}{e(X)}\right]$. It says nothing about whether you can estimate it well.

- › Both routes require a nuisance function of X : the outcome regression or the propensity score $e(x)$
- › Suppose $e(x)$ is a **really gnarly** function of X : many interactions, thresholds, regions where it sits near 0 or 1
- › Then *any* estimator of $e(x)$ converges slowly, and its errors land in the denominator of your weights
- › Unconfoundedness held all along. It did not help.

Extrapolation makes this worse: where treated and control covariate distributions barely overlap, the estimate of $e(x)$ is driven by the model, not the data.



Why experiments are different

In a randomised experiment you do not *estimate* the propensity score. You **know** it, because you chose it.

- › $e(x) = 1/2$ for every unit, or whatever the design specified
- › No nuisance function to learn, no rate of convergence to worry about, no model to misspecify
- › The Horvitz-Thompson estimator from last week is unbiased at every N *because* $e(x)$ is known

The Takeaway: the propensity score is trustworthy when it comes from a **design** and doubtful when it comes from a **model**. Balancing methods, today's topic, are what you do when you have to model it anyway.



The focus of today

The Theory vs. Practice Gap: With some assumptions, propensity score weighting should achieve perfect balance in expectation, but we observe that *finite sample balance is often poor*.

Why balance fails:

- › **Model misspecification:** Our estimated $\hat{e}(x)$ may not equal the true $e(x)$
- › **Extreme weights:** Units with very low/high propensity scores get huge weights, creating instability
- › **Finite sample variation:** Even with correct model, randomness leads to imperfect balance

What we'll explore:

- › Methods that directly optimize for covariate balance
- › The bias-variance trade-off in balancing weights

Key insight: We can target balance directly rather try to get the “right” propensity score.



(Imai and Ratkovic 2014): Covariate Balancing Propensity Score

The big picture

- › **The Problem:** The propensity score must be *estimated*, and slight misspecification of that model can produce large bias in the effect estimate.
- › **The Paradox:** The propensity score exists to reduce a high-dimensional covariate vector to one number, yet estimating it means modelling those same high-dimensional covariates.
- › **The Move:** The propensity score is *two things at once*: a conditional probability of treatment and a covariate balancing score. Use the second property to estimate the first.
- › **The Result:** Fit the same logistic model, but choose β so that the implied weights *balance the covariates*, not so that the likelihood is maximised. Estimation runs through GMM.

Imai & Ratkovic write T_i , $\pi_\beta(X_i)$ and $\tilde{X}_i$. We write A_i , $e_\beta(X_i)$ and $f(X_i)$.



The propensity score tautology

The standard workflow:

1. Fit a logistic regression $e_{\beta}(X_i) = \text{logit}^{-1}(X_i^{\top} \beta)$ by maximum likelihood
2. Check covariate balance in the weighted or matched sample
3. If balance is poor, change the specification and go back to 1

“The estimated propensity score is appropriate if it balances covariates.”

- › The model is judged by a criterion the estimator **never targeted**
- › The search is manual, unprincipled, and not reproducible
- › ([Smith and Todd 2005](#)): even mild misspecification wrecks propensity score matching estimators

The fix: make balance the estimation criterion, not a post-hoc check.



Two faces of the propensity score

Face 1: a model for treatment. The MLE first-order condition is

$$\frac{1}{N} \sum_i \left[\frac{A_i e'_\beta(X_i)}{e_\beta(X_i)} - \frac{(1 - A_i) e'_\beta(X_i)}{1 - e_\beta(X_i)} \right] = 0, \quad e'_\beta(X_i) = \frac{\partial e_\beta(X_i)}{\partial \beta}$$

Face 2: a balancing score. Inverse weighting by the *true* $e(X)$ makes any function of X mean-independent of treatment:

$$\mathbb{E} \left[\frac{A_i f(X_i)}{e_\beta(X_i)} - \frac{(1 - A_i) f(X_i)}{1 - e_\beta(X_i)} \right] = 0 \quad \text{for any } f$$

The observation: the score condition *is* a balancing condition, with $f(X_i) = e'_\beta(X_i)$. Maximum likelihood balances the one function of X the model says predicts treatment. If the model is wrong, that is the **wrong function to balance**.



The CBPS moment conditions

Pick the functions you actually want balanced and impose them as moment conditions:

$$\bar{g}_\beta = \frac{1}{N} \sum_i \frac{A_i - e_\beta(X_i)}{\underbrace{e_\beta(X_i)\{1 - e_\beta(X_i)\}}_{w_\beta(A_i, X_i)}} f(X_i) = 0$$

- › $f(X_i) = X_i$ balances **first moments** of every covariate, whether or not the logistic model is right
- › $f(X_i) = (X_i^\top, X_i^{2\top})^\top$ balances **first and second moments**
- › For the ATT, weight the controls to match the treated: $w_\beta = \frac{N}{N_1} \frac{A_i - e_\beta(X_i)}{1 - e_\beta(X_i)}$
- › **Just-identified** CBPS: balance conditions alone, as many conditions as parameters. Solve $\bar{g}_\beta = 0$ exactly.
- › **Over-identified** CBPS: stack the balance conditions on top of the score condition. More conditions than parameters: the model cannot satisfy them all.



Estimation: generalised method of moments

With more conditions than parameters, minimise a weighted quadratic form:

$$\hat{\beta}_{\text{GMM}} = \arg \min_{\beta} \bar{g}_{\beta}^{\top} \Sigma_{\beta}^{-1} \bar{g}_{\beta}, \quad \Sigma_{\beta} = \frac{1}{N} \sum_i \mathbb{E}[g_{\beta}(A_i, X_i) g_{\beta}(A_i, X_i)^{\top} | X_i]$$

- › **Continuous-updating GMM:** Σ_{β} is re-evaluated at every β , which is invariant to reparameterisation and better in small samples
- › Σ_{β} integrates out A_i given X_i . The sample covariance of the moments does not **penalise large weights**; this one does.
- › Start from $\hat{\beta}_{\text{MLE}}$ and use gradient methods; an empirical-likelihood version exists too

A specification test for free. Over-identification gives Hansen's J-statistic:

$$J = N \bar{g}_{\hat{\beta}}^{\top} \Sigma_{\hat{\beta}}^{-1} \bar{g}_{\hat{\beta}} \xrightarrow{d} \chi_{L+M}^2$$

under the null that the propensity model is correct. Failing to reject is **not** evidence of correct specification; the test can lack power.



Why LaLonde is *the* benchmark

- › ([LaLonde 1986](#)): NSW randomised job training, giving an **experimental estimate** as the target. Replace the experimental controls with CPS or PSID non-participants; the econometric estimators of the day fail to recover it.
- › ([Dehejia and Wahba 1999](#)): propensity score matching **recovers the benchmark**. Observational methods are rehabilitated, and LaLonde becomes the standard test case.
- › ([Smith and Todd 2005](#)): the recovery depends on the subsample and the propensity specification. On the full data, matching fails.
- › **Why it is useful:** real covariates, real selection into a programme, a **known answer**.
- › **The LaLonde fallacy:** one data set, hundreds of papers. Methods have been tuned to pass this test until passing it means little. Success on LaLonde is **necessary, not sufficient**.



LaLonde, revisited

Setup, following ([Smith and Todd 2005](#)):

- › Compare NSW **experimental controls** with **PSID non-participants**
- › Neither group was trained, so the true “effect” of being in the experimental sample is **zero**
- › Outcome: 1978 earnings. Any nonzero estimate is *evaluation bias*

Specification	GLM	Just-identified CBPS	Over-identified CBPS
Smith & Todd, 1-to-1 matching	-1438	-821	-641
Smith & Todd, optimal 1-to-N matching	-1230	-827	-464

- › CBPS **roughly halves** the bias, across all three propensity specifications
- › Still biased by several hundred dollars, but **stable across specifications**: better balance, bought with some likelihood



Takeaways

- › **The fundamental insight:**

- » Maximum likelihood already balances *something*: the derivative of the propensity model with respect to its parameters
- » CBPS swaps that implicit choice for the functions of X the analyst actually cares about

- › **What you get:**

- » A familiar parametric propensity score that is robust to mild misspecification
- » Weights, matches and subclasses from any downstream propensity method, unchanged
- » An over-identification test, and a bias-variance dial between just- and over-identified versions

- › **What it does not solve:**

- » Which functions to balance. First moments of X are a default, not a theory.
- » Severe misspecification. The logistic form is still imposed.

- › **The bridge:** the estimator only ever sees the treatment and covariates, never the outcome. Design stays separate from analysis.



Discussion: CBPS

1. CBPS balances the functions you chose **by construction**. The balance table in your paper will therefore look perfect. **What diagnostic is left** that could tell you the propensity model is wrong?
2. You balance the means of 40 covariates. A referee asks why not the interactions and squares too. The paper admits there is no theory for which functions to balance. **What do you say**, and what would go wrong if you balanced all of them?
3. CBPS halves the evaluation bias on LaLonde but does not remove it. Is a method that is *stably wrong* across specifications better than one that is *sometimes right*?



(Wang and Zubizarreta 2020): Minimal Weights



The big picture

- › **The Problem:** IPW balances covariates only *in expectation*. In any one data set, especially a small one with poor overlap, balance can be bad and a few units can carry enormous weights.
- › **The Move:** Skip the propensity model. Solve for the weights *directly*: the least dispersed weights that bring the covariates into (approximate) balance.
- › **The Result:** These weights are secretly a shrinkage estimate of $1/e(x)$. That connection is what makes them consistent for the true inverse probability weights.
- › **Why it matters:** *Approximate* balance trades bias for variance. Exact balance cannot, and sometimes has no solution at all.

Wang & Zubizarreta write Z_i , $\pi(x)$ and a missing-data problem. We write A_i , $e(x)$ and think of estimating $E[Y(1)]$ from the treated arm.



Minimal weights as a mathematical program

Estimate $E[Y(1)]$ with the linear estimator $\hat{Y}_w = \sum_i w_i A_i Y_i$, choosing the weights by

$$\begin{aligned} & \underset{w}{\text{minimize}} && \sum_i A_i f(w_i) \\ & \text{subject to} && \left| \sum_i w_i A_i B_k(X_i) - \frac{1}{n} \sum_i B_k(X_i) \right| \leq \delta_k, \quad k = 1, \dots, K \\ & && \sum_i w_i = 1, \quad w_i \geq 0 \end{aligned}$$

- › f is a **convex dispersion measure** of the weights: variance, absolute deviation, negative entropy
- › $B_1, \dots, B_K$ are **basis functions** of the covariates: moments, splines, wavelets
- › $\delta_k \geq 0$ is the **tolerated imbalance** in the k th function. $\delta = 0$ is *exact* balance.
- › Sum-to-one and positivity together give $0 \leq w_i \leq 1$: the weights **never extrapolate**



One program, many methods

Method	$f(w)$	δ
Entropy balancing (Hainmueller 2012)	$w \log w$	0
Stable balancing weights (Zubizarreta 2015)	$(w - 1/r)^2$	≥ 0
Empirical balancing calibration (Chan, Yam, and Zhang 2016)	$D(w, 1)$, a strictly convex distance	0

- › Most methods balance **exactly**. This paper studies the whole class, $\delta \geq 0$.
- › Tighten δ : better balance, **extreme weights**, possibly **infeasible**.
- › Loosen δ : stable weights, **residual imbalance**.
- › **The question**: what does that dial actually control?



Deriving the dual I: the Lagrangian

Drop the sum-to-one and positivity constraints for clarity, and write the imbalance vector

$$g(w) = \sum_i A_i w_i B(X_i) - \frac{1}{n} \sum_i B(X_i), \quad \text{constraint: } -\delta \leq g(w) \leq \delta$$

Two-sided constraints get two multiplier vectors, $\lambda^+, \lambda^- \geq 0$:

$$\mathcal{L} = \sum_i A_i f(w_i) + (\lambda^+)^T \{g(w) - \delta\} - (\lambda^-)^T \{g(w) + \delta\}$$

Set $\lambda = \lambda^+ - \lambda^-$, now **free in sign**. For a given λ the dual is largest when one of the pair is zero, so $\lambda^+ + \lambda^- = |\lambda|$:

$$\mathcal{L}(w, \lambda) = \sum_i A_i \left[f(w_i) + w_i B(X_i)^T \lambda \right] - \frac{1}{n} \sum_i B(X_i)^T \lambda - |\lambda|^T \delta$$

- › The box constraint has become an ℓ_1 **term** in λ . That is where the Lasso will come from.
- › Each w_i can be minimised on its own.



Deriving the dual II: solve for the weights

Minimise over each treated unit's weight. The first-order condition and its solution are

$$f'(w_i) = -B(X_i)^\top \lambda \implies w_i^* = (f')^{-1}\{-B(X_i)^\top \lambda\} =: \rho'\{B(X_i)^\top \lambda\}$$

Substitute back and flip the sign of the maximisation:

$$\underset{\lambda}{\text{minimize}} \quad - \sum_i A_i \rho\{B(X_i)^\top \lambda\} + \frac{1}{n} \sum_i B(X_i)^\top \lambda + |\lambda|^\top \delta \quad (\text{Theorem 1})$$

› Convex program, so **strong duality**:

» solve for λ^+ (K unknowns, not n), read off $w_i^* = \rho'\{B(X_i)^\top \lambda^+\}$

$f(w)$	$f'(w)$	$\rho'(t) = (f')^{-1}(-t)$	Implied model for w
$w \log w$ (entropy)	$\log w + 1$	$\exp(-t - 1)$	log-linear in $B(x)$
$w^2/2$ (variance)	w	$-t$	linear in $B(x)$



Balancing is shrinkage estimation of $1/e(x)$

$$\underset{\lambda}{\text{minimize}} \quad \underbrace{- \sum_i A_i \rho\{B(X_i)^\top \lambda\} + \frac{1}{n} \sum_i B(X_i)^\top \lambda}_{\text{loss}} + \underbrace{|\lambda|^\top \delta}_{\text{penalty}}$$

- › **The loss is a GLM.** Its population first-order condition at $\delta = 0$ is $E[A \rho'\{B(X)^\top \lambda\} B(X)] = E[B(X)]$, which $nw^*(x) = 1/e(x)$ satisfies exactly. The weights are a **model for the inverse propensity score** with link ρ' and coefficients λ .
- › **The penalty is a weighted Lasso.** λ_k is the **shadow price** of the k th balance constraint. Functions that are hard to balance get their coefficient **shrunk to zero** and their constraint goes slack.
- › So δ is a **regularisation parameter**, and exact balance is the **unregularised** fit.



Why $1/e(x)$? The Riesz representation

The target is a **linear functional** of the outcome regression $\mu(x) = E[Y | X = x, A = 1]$:

$$\psi = E[Y(1)] = E[\mu(X)]$$

Riesz: every bounded linear functional on L_2 is an inner product. So there is an $\alpha(X)$:

$$E[\mu(X)] = E[A \alpha(X) \mu(X)] = E[A \alpha(X) Y] \quad \text{for every } \mu$$

Solving $E[e(X) \alpha(X) h(X)] = E[h(X)]$ for all h gives $\alpha(X) = 1/e(X)$. The inverse propensity is the **Riesz representer** of “average the regression.”

- › It provides a **balance condition on every function** h .
 - » Balancing weights impose it on $h \in \text{span}\{B_1, \dots, B_K\}$.
- › With $f(w) = w^2/2$ and $\alpha_i = nw_i$, the dual loss is $\frac{1}{n} \sum_i [A_i \alpha(X_i)^2 - 2 \alpha(X_i)]$
 - » **least-squares regression for α** with no observed outcome.
 - » Its population minimiser is $\alpha = 1/e$.
- › Two routes to the same object: model $e(x)$ and invert it, or regress the representer directly. Balancing weights take the second.



Consistency: the weights estimate $1/e(x)$

The assumption. The thing the dual is fitting, $m^*(x) = (\rho')^{-1}\{1/(n e(x))\}$, is *smooth enough* that K basis functions approximate it with error $O(K^{-r_\pi})$. For splines, $r_\pi = s/d$: derivatives over dimensions.

Theorem 2. With $K = o(n)$ and δ shrinking fast enough,

$$\sup_x \left| n w^*(x) - \frac{1}{e(x)} \right| = O_p \left\{ \underbrace{\frac{K \log K}{n}}_{\text{estimation}} + \underbrace{K^{1-r_\pi}}_{\text{approximation}} \right\} = o_p(1)$$

- › **Two familiar errors.** More basis functions: more coefficients to estimate (first term grows in K), richer approximation (second shrinks). Bias-variance, now in the weights.
- › **What it says.** The weights converge *uniformly* to the true inverse probability weights, without a propensity model ever being written down.
- › **The catch.** δ must shrink with n : $\|\delta\|_2 = O_p\{K^{1/2} \log K/n + K^{1/2-r_\pi}\}$. Approximate balance, **but not too approximate**.
- › The paper continues to normality and efficiency.



How many functions should I balance?

The worry. Every balanced function is a coefficient in the implicit propensity model. Balance 200 functions, fit a 200-parameter model. Does the noise wreck the weights?

The comparison. An *oracle* knows which S functions matter and balances only those, exactly. You do not, so you balance everything approximately.

Theorem 4. Measuring quality by how far the dual loss L sits above its best possible value,

$$\underbrace{E\{L(\lambda^\dagger) - L(\lambda_1^*)\}}_{\text{your loss, balancing everything}} \leq 3 \times \underbrace{E\{L(\lambda^\ddagger) - L(\lambda_1^*)\}}_{\text{the oracle's loss}} + c_0 S$$

- › **In words:** at most three times the oracle's loss, plus a fixed price per function the oracle needed. Not knowing costs little.
- › **Why:** the ℓ_1 penalty zeroes most coefficients, so few constraints bind. It does the oracle's selection for you.
- › **In practice:** balance generously and let δ decide which constraints matter.



How do I choose δ ?

There are no outcomes at the design stage, so no obvious out-of-sample target. Their proposal:

Algorithm 1. For each δ on a grid in $[0, K^{-1/2}]$

1. Solve the program on the original data to get $w(\delta)$
2. Draw bootstrap samples; on each, compute the **standardised imbalance** of every B_k using the *original* weights $w(\delta)$
3. Average the imbalance over bootstrap samples and functions

Return the δ with the **smallest bootstrap imbalance**.

- › **Why it works:** the true inverse-propensity weights balance the population, so they also balance any sample from it. Weights that balance the sample but *not* its resamples are overfit.
- › The cap $K^{-1/2}$ keeps δ inside the range Theorem 2 allows.



Takeaways

- › **The fundamental insight:**

- » Weights that minimise dispersion subject to balance constraints are an ℓ_1 -regularised model for $1/e(x)$
- » The tolerance δ is the regularisation parameter, and exact balance is the unregularised fit

- › **What the theory buys:**

- » Consistency for the true inverse probability weights, and an interpretation: they estimate the Riesz representer of the target
- » Balancing too many functions is safe: the loss is bounded by an oracle inequality

- › **What to do in practice:**

- » Balance approximately, not exactly, and tune δ by bootstrap balance
- » In their simulations, gains appear when overlap is poor: exact balance is often infeasible there, and approximate balance can halve the RMSE

- › **What it does not do:**

- » Guard against an outcome that depends on functions of X you did not put in B



Discussion: Minimal Weights

1. Exact balance feels like the *safe* choice: no residual imbalance to explain away. The paper says it is the **least regularised** choice. Which intuition wins, and when?
2. Rosenbaum and Rubin tell us to model the treatment; minimal weights say to never fit that model explicitly. Theorem 1 says you fit it **anyway**. Is anything gained by hiding the propensity score inside a dual problem?
3. Algorithm 1 tunes δ without looking at outcomes. That keeps design and analysis separate. **What are you giving up** by refusing to use outcome information here?



More to read

Paper	Why you'd read it
(Ben-Michael et al. 2021)	The survey that frames today: modelling versus balancing as two routes to the same weights, and balance as a <i>sufficient condition</i> for robust inference.
(Zhao 2019)	The theory behind CBPS: every link and estimand defines a scoring rule whose optimum balances covariates, at no asymptotic efficiency cost.
(Arbour, Dimmery, and Sondhi 2021)	Balancing as classification: distinguish the data from a permuted copy and the odds are the weights. Continuous treatments come free.
(Hainmueller 2012) ; (Zubizarreta 2015)	The two originals behind the minimal-weights table: exact entropy balancing, and variance-minimising weights with tolerance bands.
(Li, Thomas, and Li 2019)	When overlap is poor and you would rather change the estimand than trim: weight toward the population where treatment is a coin flip.



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